

AI Agents at Every Scale

Personal, Team, and
Data Workflows

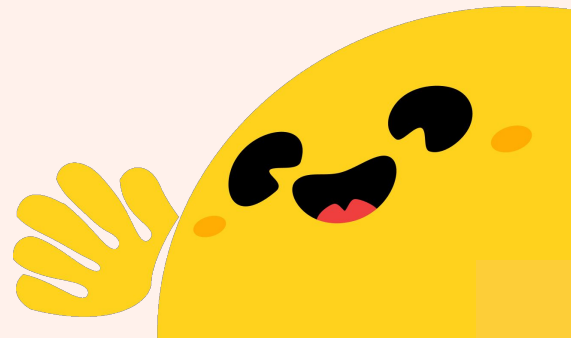
@  **Hugging Face**



01

Introduction

Let's dive in



Who am I?



Quentin Lhoest

 lhoestq  lhoestq



AI Data Engineer

Focused on bringing the power of AI to everyone



Joined Hugging Face in 2020

On a mission to democratize machine learning and NLP (aka AI)



Passionate about Open Source and Data

Creator of the `datasets` library

HF Datasets (1M+ public datasets)

HF Buckets (100+ PB, 80% deduped thx to Xet & Parquet deduplication)



100%
Of people at HF
use AI agents

Here is what we do

02

Personal AI

The rise of individual agents



The agent is a personal tool



1- Harness

What interface ? Which tools and skills ?

2- Model

What expertise ? Which modalities ?

3- Provider

What cost and latency ? What about my data ?
How to track usage and costs ?

-> HF Datasets for agent traces + HF inference for tracking



What we use at 🤗

Harnesses

CLI



Claude Code



Codex CLI



Pi



Open Code

IDE



Codex



VS Code w/ codex



Cursor



Claude desktop

...or web

~most common

~less common

Models

Closed



Claude



GPT

Open



Kimi



GLM



Deepseek



Qwen



Gemma

What we use at

Harnesses

CLI

 Claude Code

 Codex CLI

 Pi

 OpenAI



IDE

 Codex

 VS Code w/ codex

 Cursor

 Claude desktop

...or web

Closed

 Claude

 GPT

Models

Open

 Kimi

 GLM

 Deepseek

 Qwen

 Gemma



most common

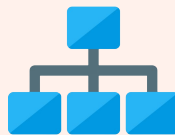
less common

What is ?



Open Source harness

Modern tool for agents in
tech environments



Agnostic

Compatibility with any
provider or local server,
easy to switch



General-purpose

Minimal and tweakable harness
with plugins



*There are many agent harnesses
but this one is yours*

Pi is a minimal agent harness.
Adapt Pi to your workflows, not the other way around.



Say hello to the Open Lakehouse + AI Meetup !

The user wants me to "Say hello to the Open Lakehouse + AI Meetup!" This is a fun, light request. Maybe they want a greeting message. Let me just say hello in a friendly way, perhaps with some flair about lakehouses and AI.

No tools needed really. Just respond.

Maybe I should check if there's something in the working directory related to a meetup? The request seems casual. I'll just craft a greeting.

Keep it concise and fun.

👋 Hello, Open Lakehouse + AI Meetup!

(raises a mug of coffee in a well-lit conference room)

Here's to the people who believe data should be open, tables should be typed, and nobody – nobody – should have to register a UDF just to read an S3 path.

May your catalogs be unified, your schema evolution painless, your compactions fast, and your `SELECT` * cheap. 🍷📊✨

Cheers! 🍷

|



Open Models for Managed & Local Inference



Security & Privacy

- No sensitive data ever sent to a remote server
- Define your own guardrails



Customized

- Many levels of expertise
- Fine-tune to your needs
- Dedicated



Lower costs

- Open-weight models are cheaper to run
- Use efficient hardware
- Optimize cost/speed/intelligence



Local AI is scalable

- Distribution over the individuals in the company



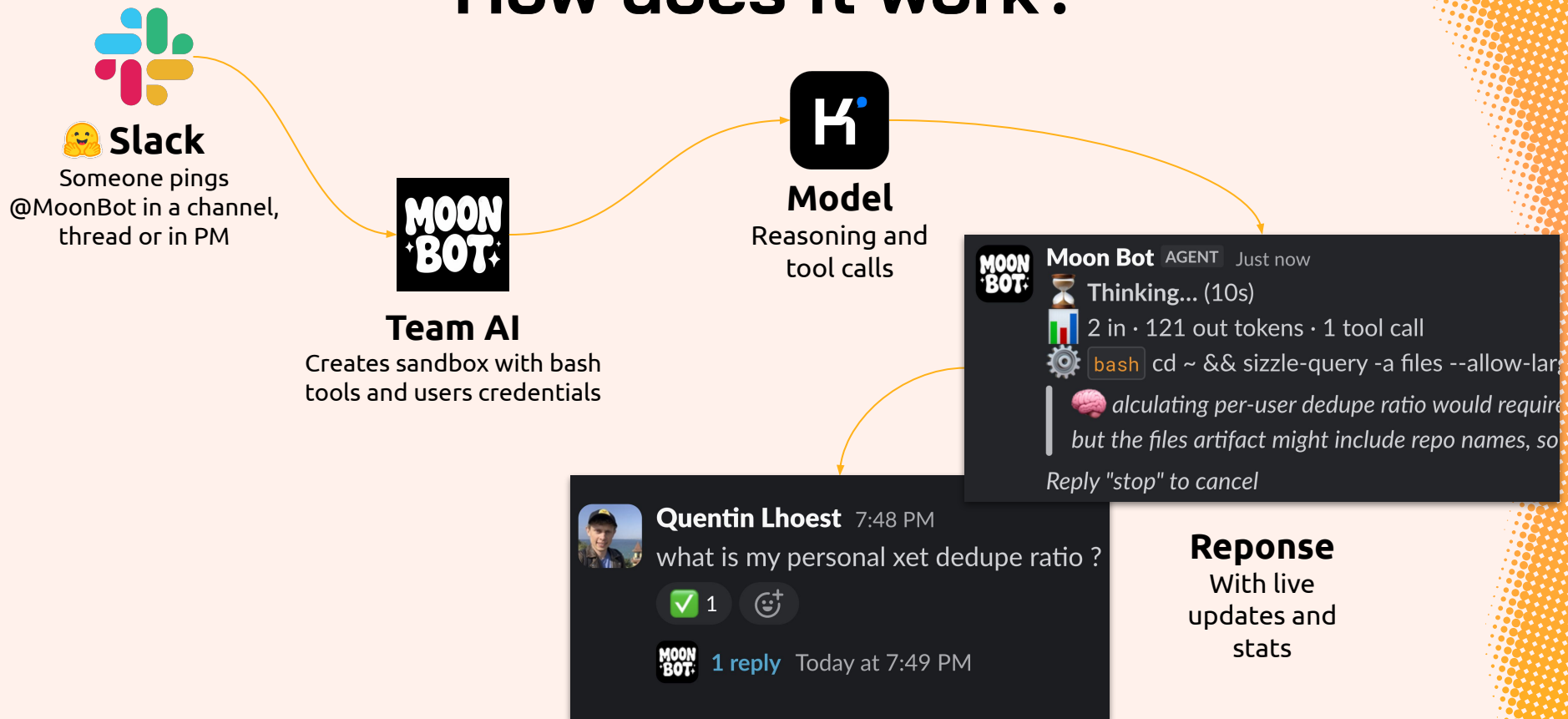
03

AI for Teams

The perfect HF member assistant



How does it work?

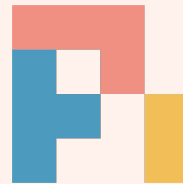


Inside Moon Bot

01

Load a new Pi session

Get the context from Slack and the list of tools allowed for this user



02

Launch a sandbox (separate linux users)

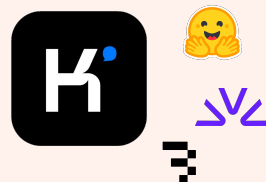
Create a working directory for the agent and prepare the credentials for this user



03

Infer with Inference Providers

Call Kimi-K3 or other models and live-update the response on Slack



Main use cases



21%

**Support and
customer
enablement**

14%

**Engineering Q&A
& code search**

14%

**Prod data
queries**

11%

**Ops runbooks &
privacy requests**

11%

**Incidents &
performance**

11%

**Code review &
shipping fixes**

7%

**Trust & safety /
abuse**



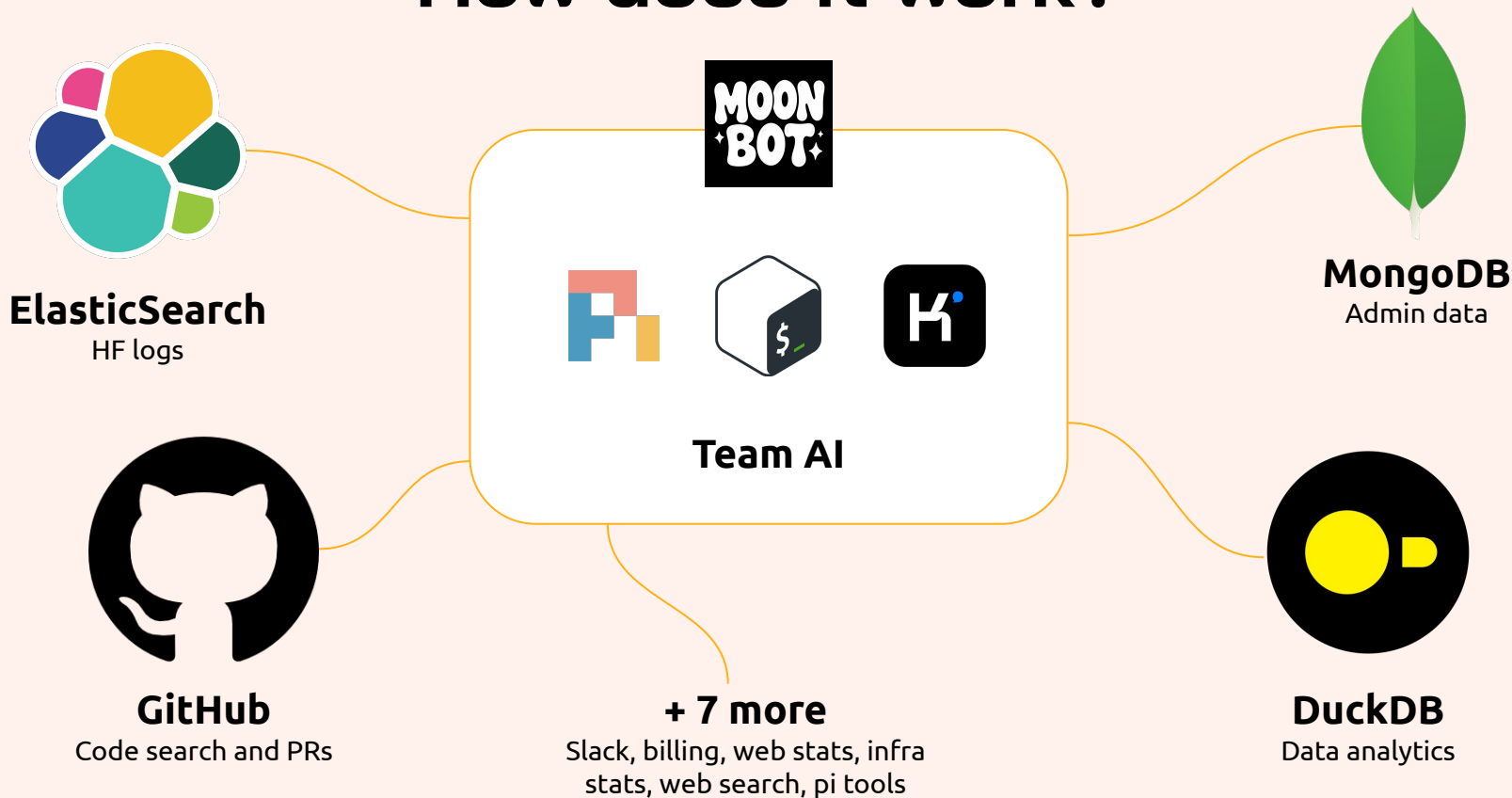
04

Data workflows

Superpowers for our Team AI



How does it work?



How does it work?



ElasticSearch

HF logs



GitHub

Code search and PRs



SKILL.md

- List of commands
- Forensics tips

es-cli + proxy

- Injects the real API key in http queries
- Agent never sees the key

+ 7 more

Slack, billing, web stats, infra
stats, web search, pi tools



MongoDB

Admin data



DuckDB

Data analytics

How does it work?



ElasticSearch
HF logs



GitHub
Code search and PRs



SKILL.md

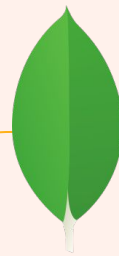
- List of databases and fields
- Common operations
- Best practices

mongosh wrapper

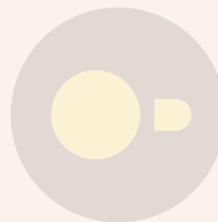
- Inject environment variables with keys
- Agent never sees the keys

+ 7 more

Slack, billing, web stats, infra
stats, web search, pi tools



MongoDB
Admin data



DuckDB
Data analytics

How does it work?



ElasticSearch
HF logs



GitHub
Code search and PRs



SKILL.md

- List of repositories and local copies
- Rules and deploy scripts

gh wrapper

- Inject environment variables with keys
- Agent never sees the keys

+ 7 more

Slack, billing, web stats, infra
stats, web search, pi tools



MongoDB
Admin data



DuckDB
Data analytics

How does it work?



ElasticSearch
HF logs



GitHub
Code search and PRs



SKILL.md

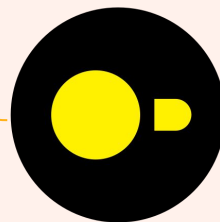
- List of tables and schemas
- Query examples

sizzle-cli + proxy

- Injects the real API key in http queries to Hugging Face datasets and buckets
- Agent never sees the key



MongoDB
Admin data



DuckDB
Data analytics

+ 7 more

Slack, billing, web stats, infra
stats, web search, pi tools

How does it work?



ElasticSearch
HF logs



GitHub
Code search and PRs



SKILL.md

- List of tables and schemas
- Query examples

sizzle-cli + proxy

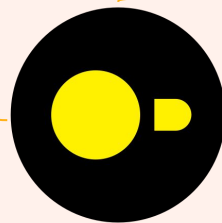
- Injects the real API key in http queries to Hugging Face datasets and buckets
- Agent never sees the key

+ 7 more

Slack, billing, web stats, infra
stats, web search, pi tools

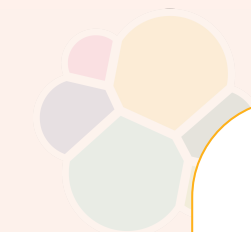


DuckLake



DuckDB
Data analytics

How does it work?



ElasticSearch
HF logs



GitHub

Code search and PRs

MOON



Hugging Face
Storage Buckets

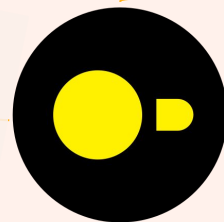
- Store data, but also models or artifacts
- Xet deduplication (80% in total !)
- Built-in CDN

+ 7 more

Slack, billing, web stats, infra
stats, web search, pi tools

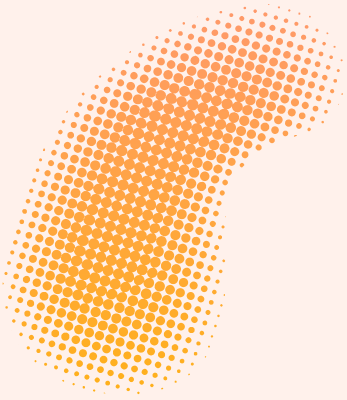


DuckLake



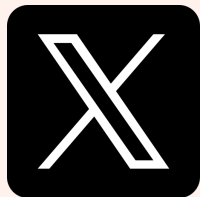
DuckDB

Data analytics



THANKS!

Let me know if you have questions :)



@lhoestq



@lhoestq

CREDITS: The presentation template was created by **Slidesgo**, and includes graphics by **Chun Te Lee**, icons by **Flaticon**, and infographics & images by **Freepik**

